

EE 635 Digital Integrated Circuit Design

Final Report for Optimised 4 Bit MULTIPLIER Project

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1. Transmission Gate XOR Gate: Schematic, Voltage Transfer Characteristics, and Propagation Delay

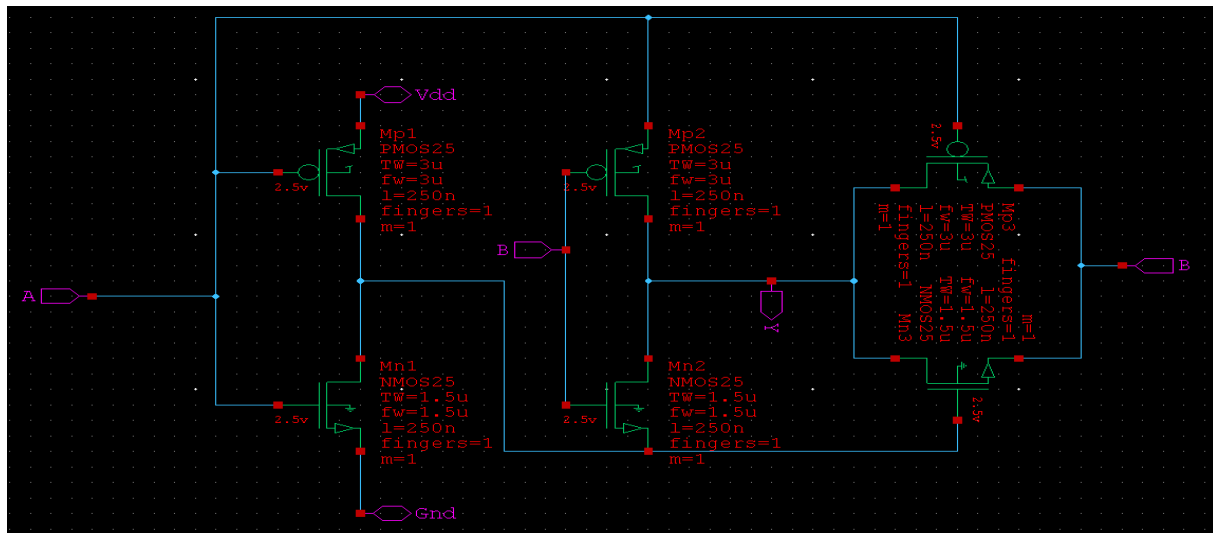


Fig. 1: TGXOR Gate Schematic to reduce Propagation delay and no. of transistors

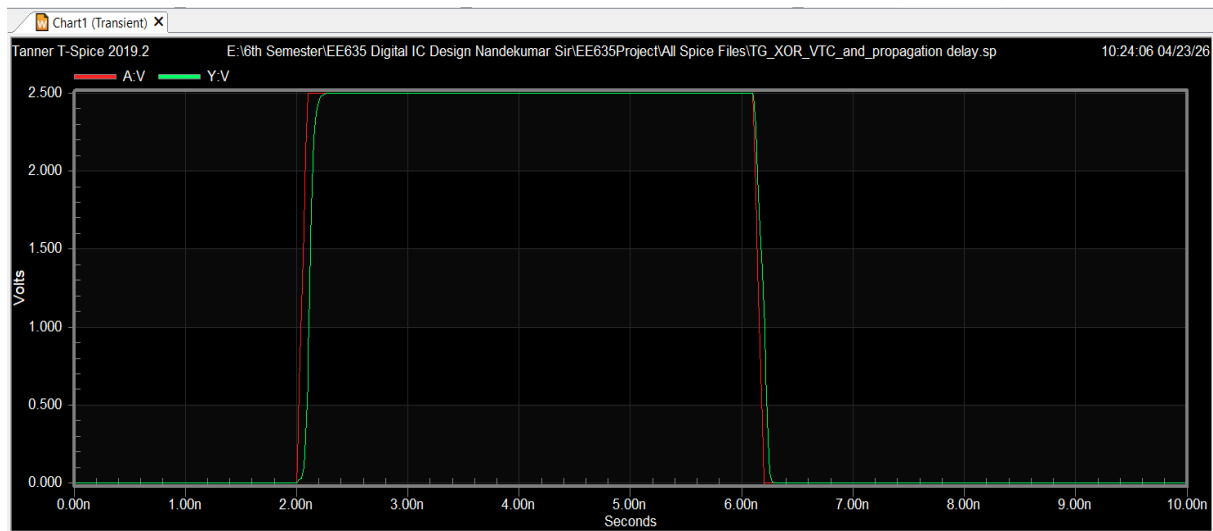


Fig. 2: TGXOR Gate VTC and Propagation delay visualisation

Propagation Delay for TGXOR Gate:

$t_{plh_xor} = 65.6759\text{psec}$

$t_{phl_xor} = 41.0952\text{psec}$

$t_{pd_xor} = 53.3855\text{psec}$

2. HALF ADDER: Schematic, Voltage Transfer Characteristics, and Propagation Delay

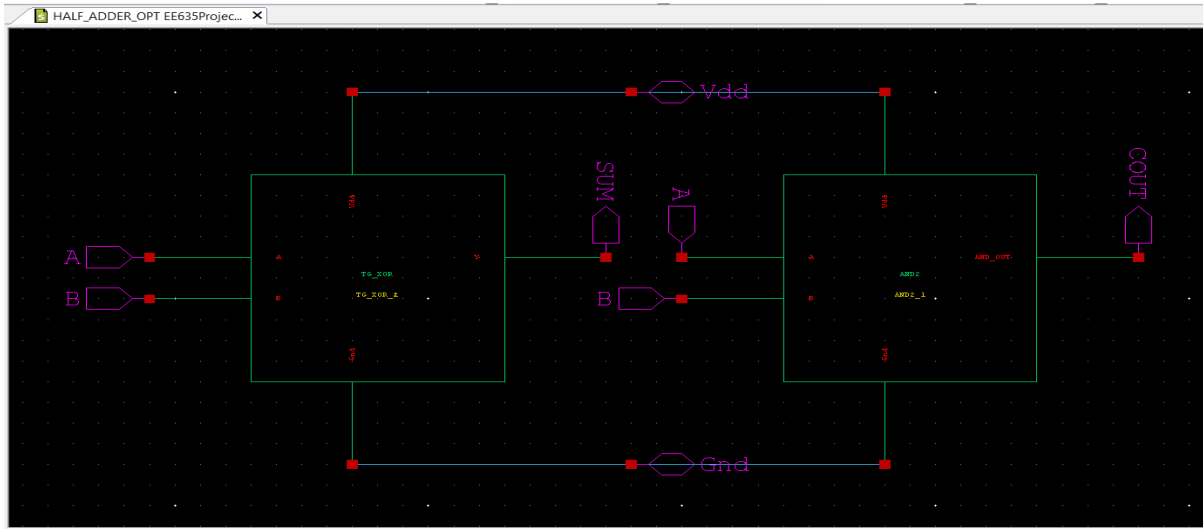


Fig. 3: Schematic for HALF ADDER using TGXOR and AND Gate.

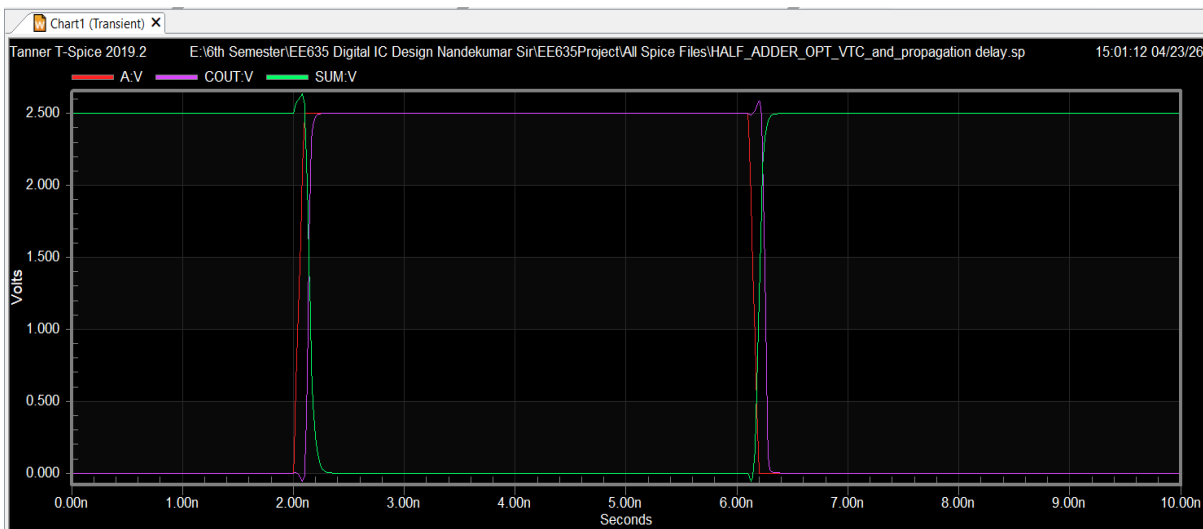


Fig. 4: HALF ADDER VTC and Propagation delay visualisation

Propagation Delay for HALF ADDER:

tphl_sum = 97.5012p
tphl_sum = 50.6677p
tpd_sum = 74.0845p
tphl_cout = 84.3643p
tphl_cout = 105.2953p
tpd_cout = 94.8298p

3. OPTIMISED FULL ADDER: Schematic, Voltage Transfer Characteristic and Propagation Delay

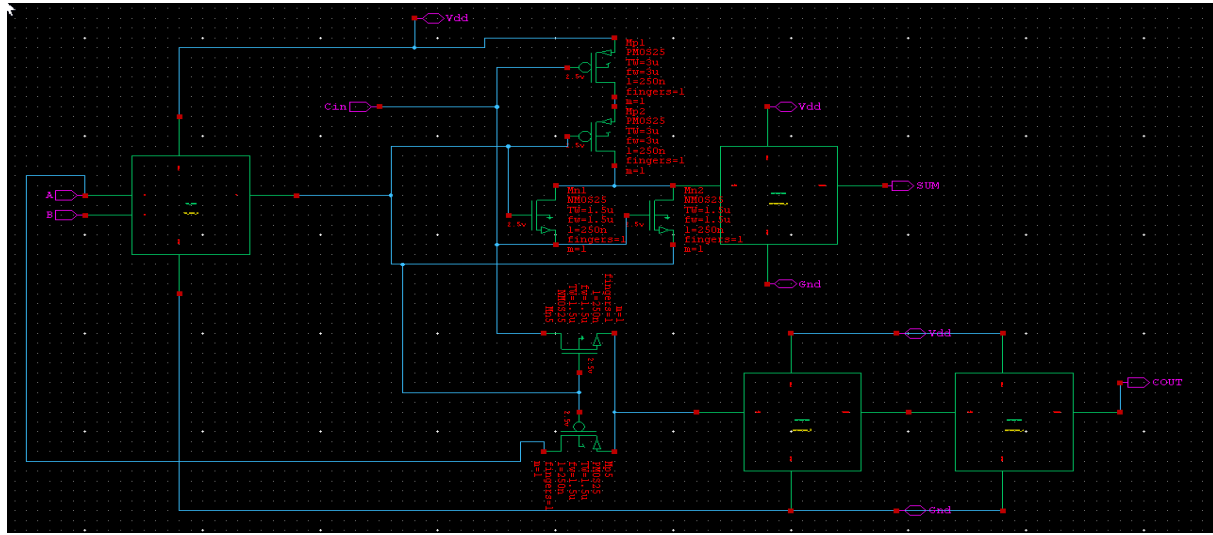


Fig. 5: Schematic for FULL ADDER with TGXOR and TSERF method of Schematic formation.

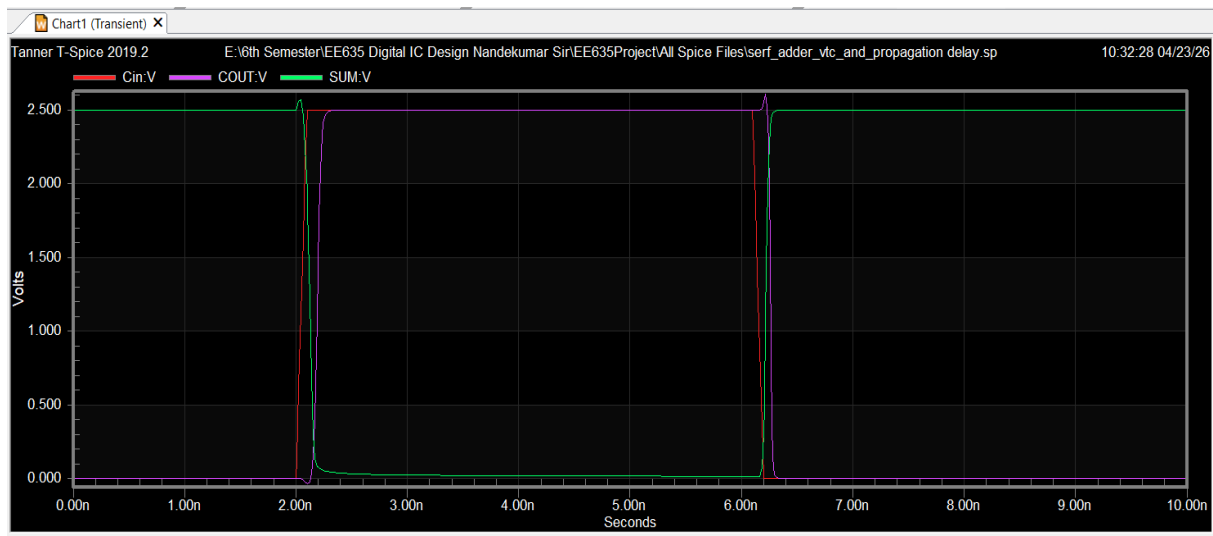


Fig. 6: Modified FULL ADDER VTC and Propagation delay visualisation

Propagation Delay for Modified FULL ADDER:

$t_{ph_sum} = 73.9941p$
 $t_{plh_sum} = 70.1498p$
 $t_{pd_sum} = 72.0720p$
 $t_{plh_cout} = 145.1781p$
 $t_{ph_cout} = 112.0969p$
 $t_{pd_cout} = 128.6375p$

4. OPTIMISED Ripple ADDER using HA and FA: Schematic, Voltage Transfer Characteristic and Propagation Delay:

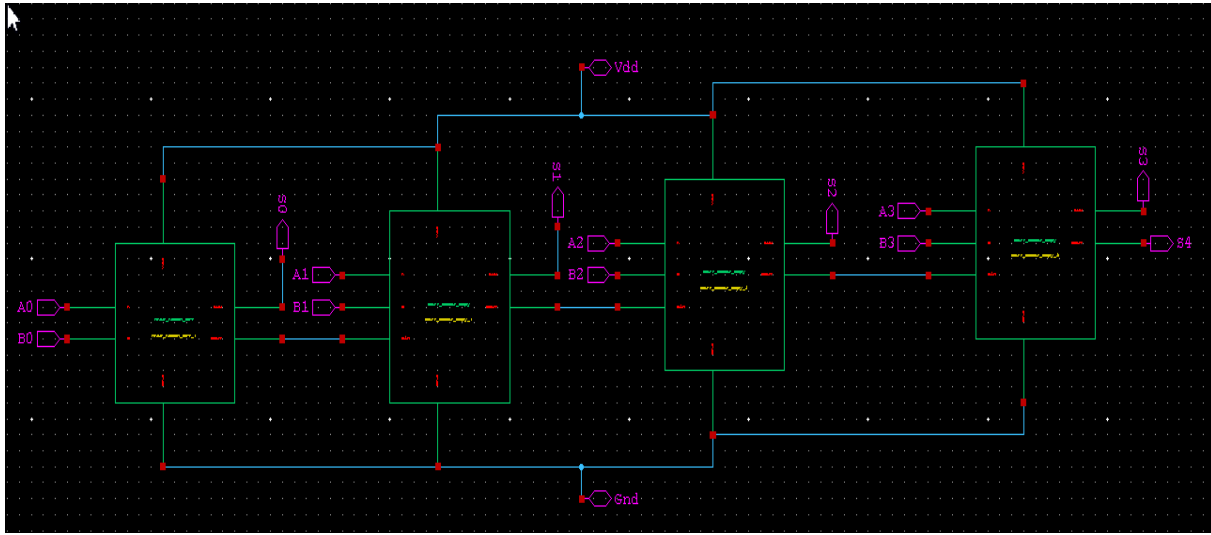


Fig. 7: Schematic for Ripple ADDER using 1 HA and 3 FA.

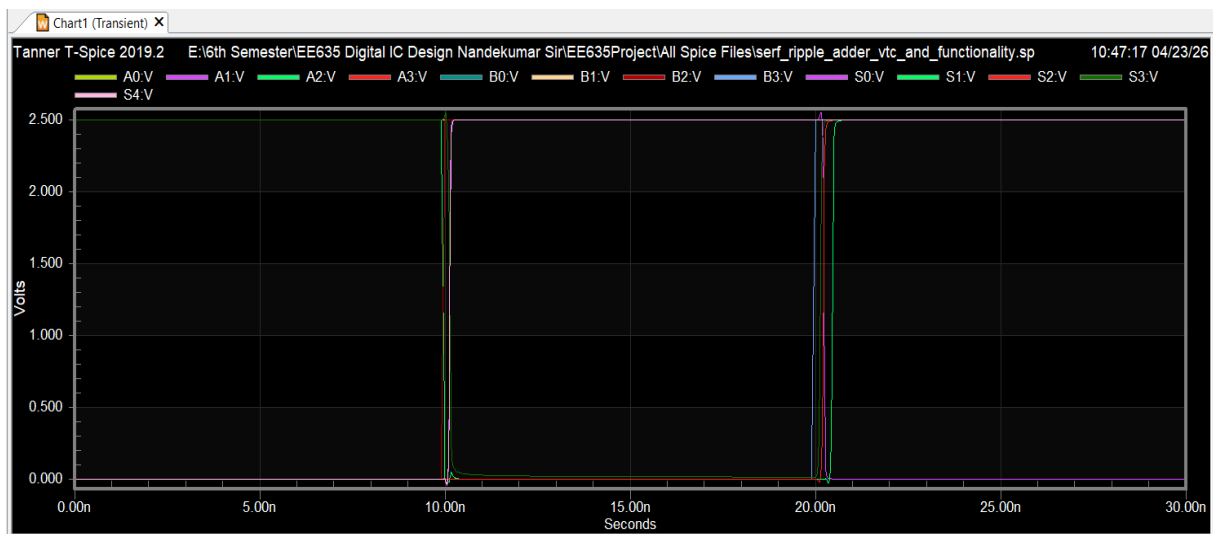


Fig. 8: Functionality of Ripple Adder.

Explanation for Waveforms:

Time Window	Decimal Operation	Input A	Input B	Expected Output
0ns – 10ns	$5 + 3 = 8$	101	11	1000
10ns – 20ns	$10 + 7 = 17$	1010	111	10001
20ns – 30ns	$15 + 15 = 30$	1111	1111	11110

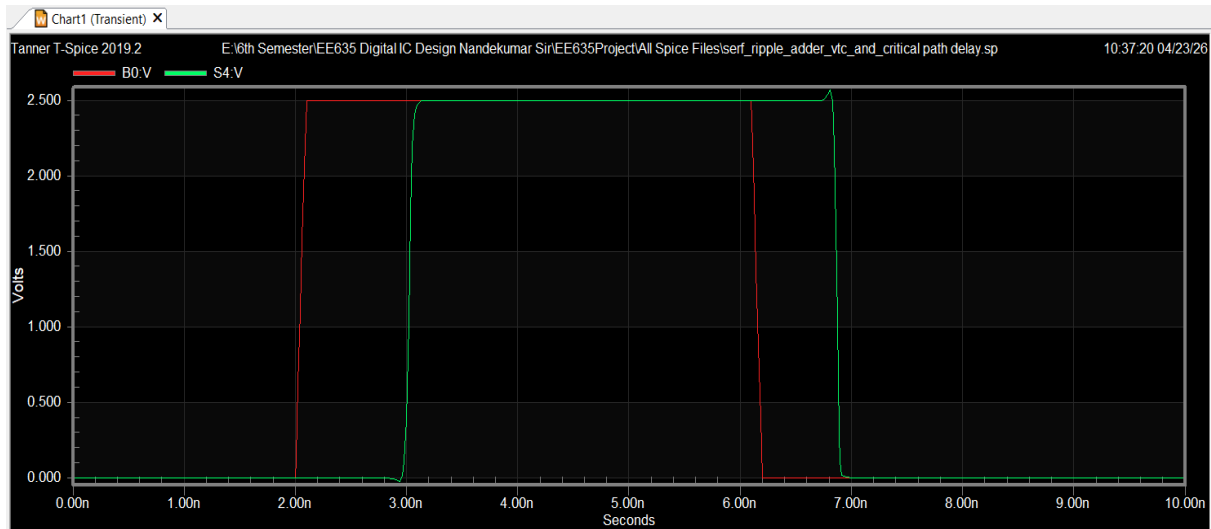


Fig. 9: Critical Path for Ripple ADDER testing

Explanation:

The critical path of a Ripple Carry Adder is defined by its worst-case propagation delay: the maximum time required for a signal to travel from the Least Significant Bit (LSB) to the Most Significant Bit (MSB).

The inputs were held at $A[3:0] = 1111$ and $B[3:1] = 000$. The LSB B_0 was then pulsed from 0 to 1. This specific transition forces a carry bit to be generated at Bit 0 and physically ripple through every subsequent adder stage until it finally toggles carry-out S_4 .

Critical Path Delay for Ripple ADDER:

$t_{plh_ripple} = 974.7759p$

$t_{phl_ripple} = 720.9045p$

$t_{pd_4bit_critical} = 847.8402p$

5. OPTIMISED MULTIPLIER using the optimised 4-bit Ripple ADDER and CMOS Static AND Gate: Schematic, Functionality and Critical Path Delay

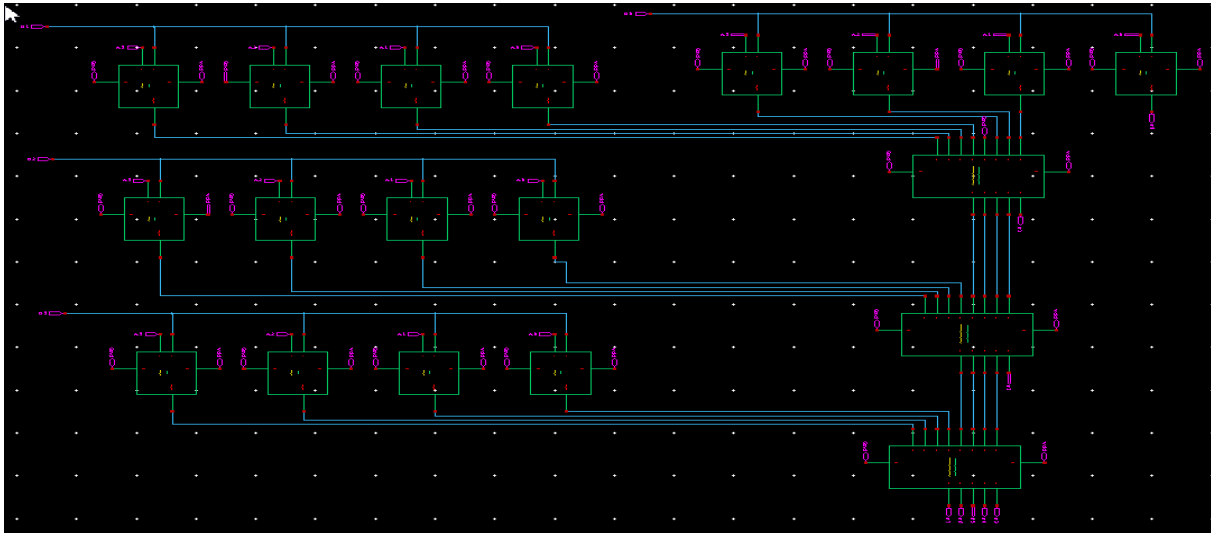


Fig. 10: Multiplier schematic using Ripple Adder and CMOS AND Gates

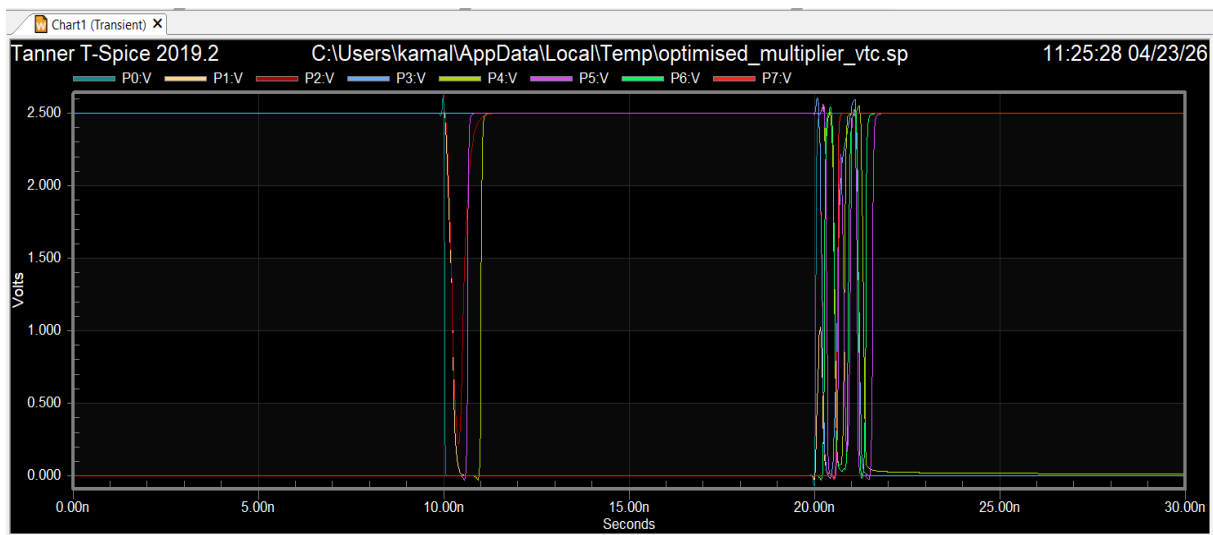


Fig. 11: Functionality of Multiplier

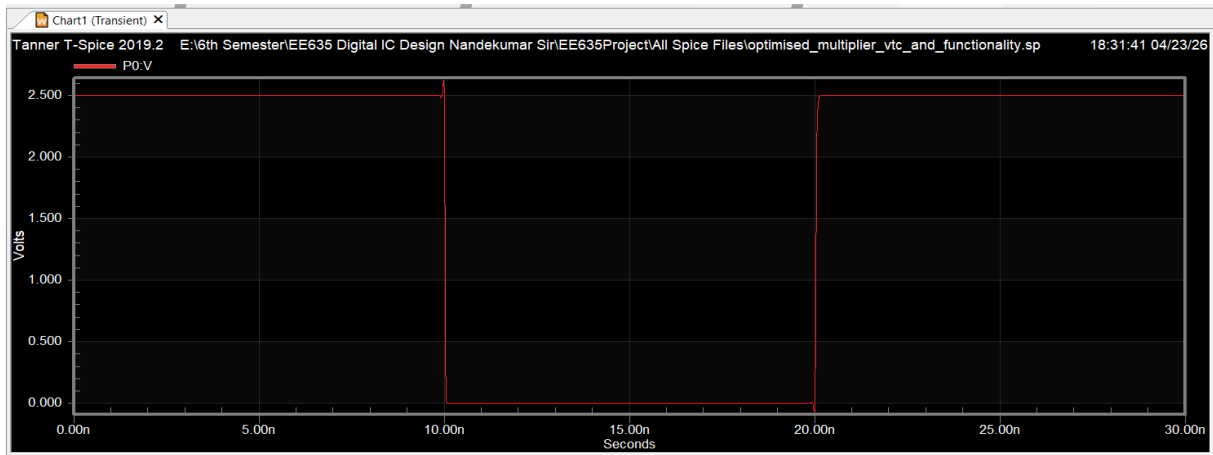


Fig. 11.1: Functionality of Multiplier: P0 plot

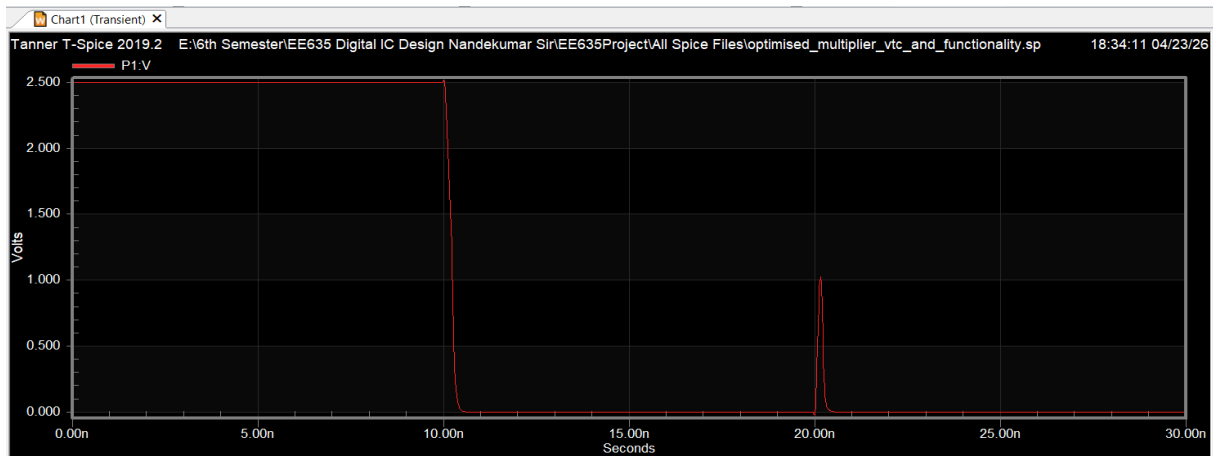


Fig. 11.2: Functionality of Multiplier: P1 plot

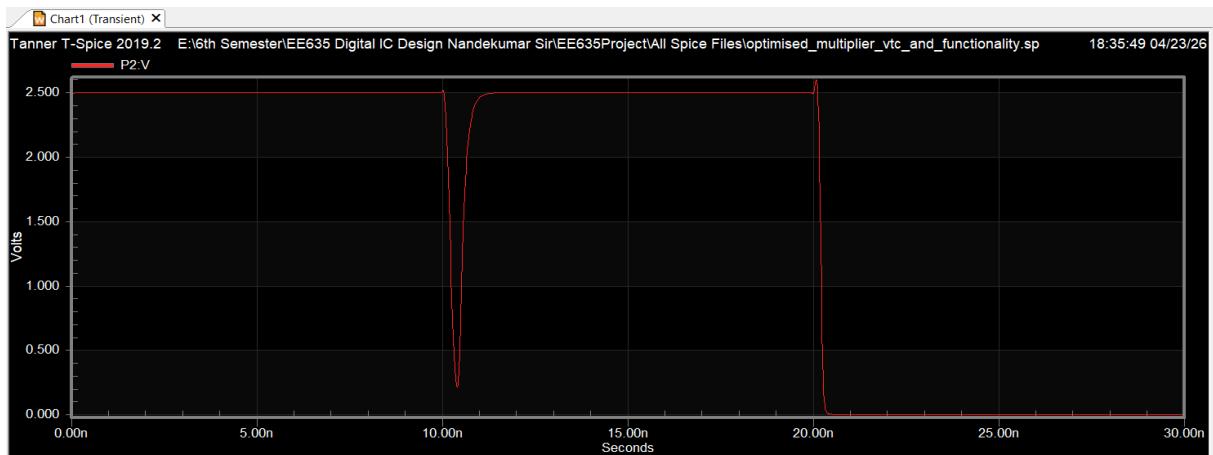


Fig. 11.3: Functionality of Multiplier: P2 plot

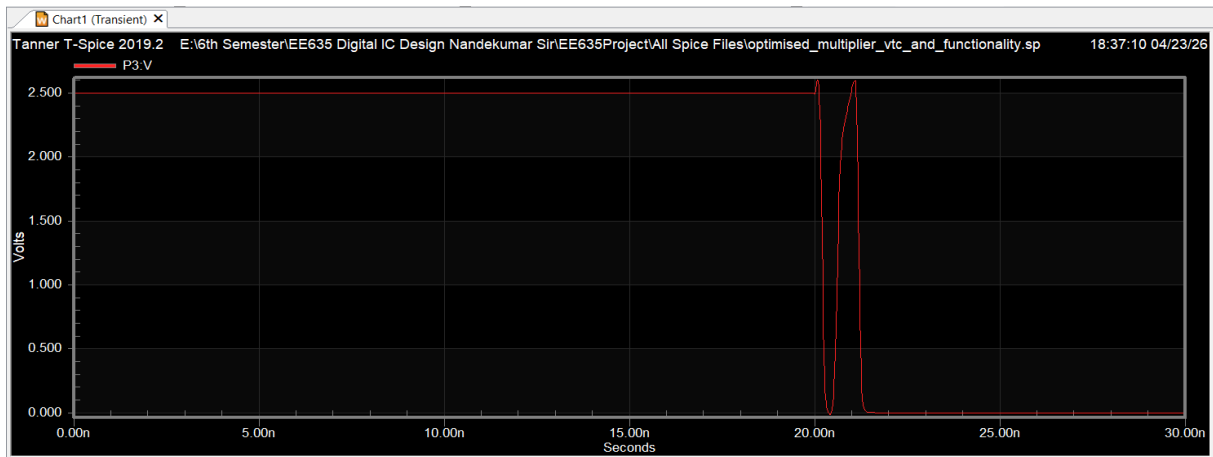


Fig. 11.4: Functionality of Multiplier: P3 plot

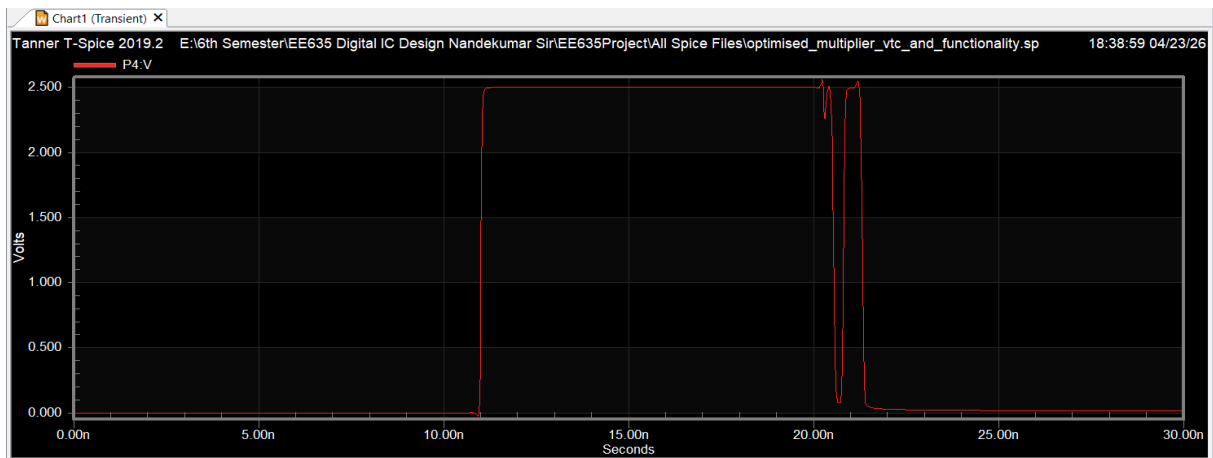


Fig. 11.5: Functionality of Multiplier: P4 plot

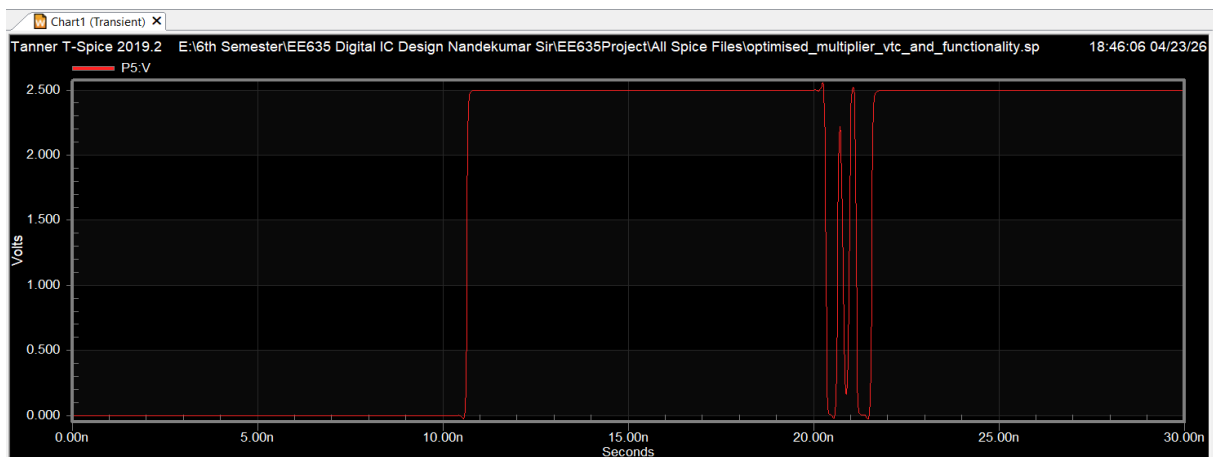


Fig. 11.6: Functionality of Multiplier: P5 plot

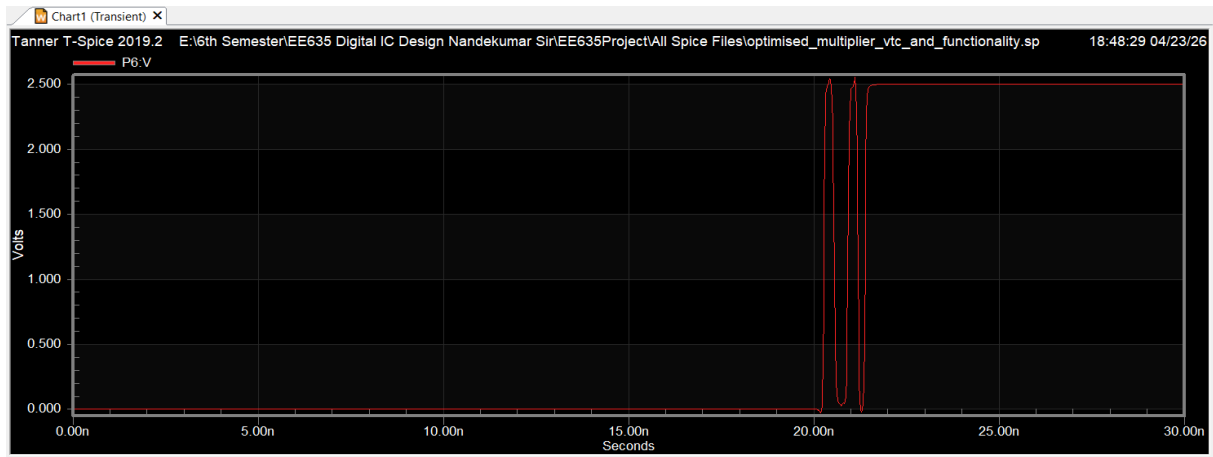


Fig. 11.7: Functionality of Multiplier: P6 plot

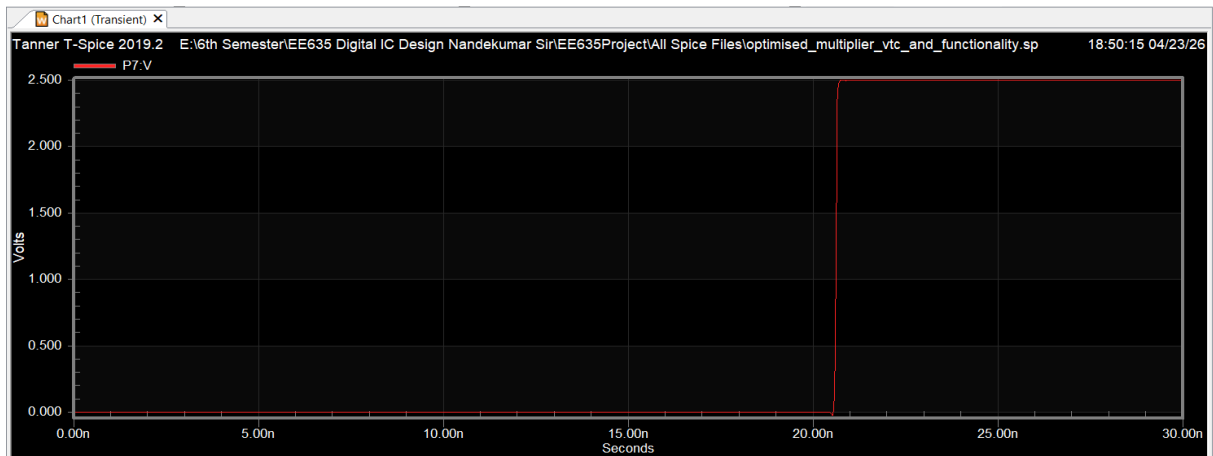


Fig. 11.8: Functionality of Multiplier: P7 plot

Explanation for Waveforms:

Time Window	Decimal Operation	Input A	Input B	Product Output
0ns – 10ns	$3 \times 5 = 15$	11	101	0000 1111
10ns – 20ns	$10 \times 6 = 60$	1010	110	0011 1100
20ns – 30ns	$15 \times 15 = 225$	1111	1111	1110 0001

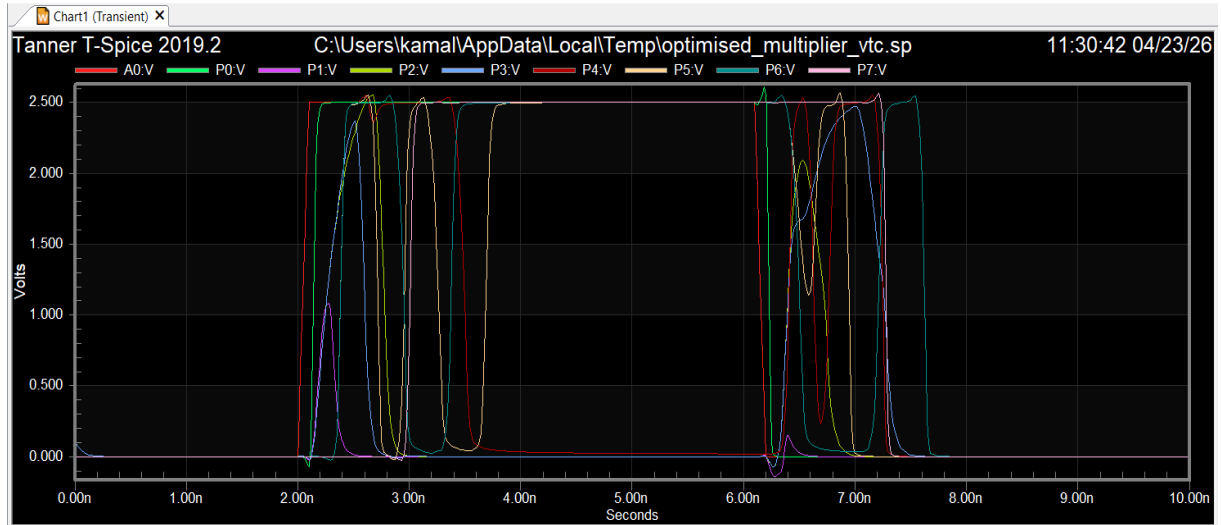


Fig. 12: Critical Path delay for the MULTIPLIER

Explanation:

The critical path delay, or maximum propagation delay of a 4x4 Carry-Save Array Multiplier, represents the longest temporal and physical route a signal must travel from the moment the input pins change state to the moment the final output bit permanently stabilises. By forcing the inputs to transition instantaneously from 0x0 to 15x15 (all input bits pulsing from Logic 0 to Logic 1), every 6T AND gate, 12T Half Adder, and 18T Full Adder is forced into simultaneous active computation. This stress test demands maximum current draw and ensures that the deepest logical paths in the matrix are fully exercised.

Critical Path Delay for MULTIPLIER:

$t_{plh_p7} = 959.5202p$

$t_{phl_p7} = 1.1291n$

$t_{pd_multiplier} = 1.0443n$

Calculation for TOTAL AREA of the MULTIPLIER:

Channel Length = $0.25\mu m$.

PMOS Width = $3\mu m$

NMOS Width = $1.5\mu m$

Area per Transistor:

PMOS: $0.25 * 3 = 0.75 \mu m^2$

NMOS: $0.25 * 1.5 = 0.375 \mu m^2$

Area per Logic Gate:

1. For CMOS AND GATE: 3PMOS + 3 NMOS
 $(3 * 0.750) + (3 * 0.375) = 3.375 \mu m^2$

2. 12T Custom Half Adder: 6PMOS + 6NMOS
 $(6 * 0.750) + (6 * 0.375) = \mathbf{6.750 \mu m^2}$
3. 18T Hybrid Full Adder: 9 PMOS + 9 NMOS
 $(9 * 0.750) + (9 * 0.375) = \mathbf{10.125 \mu m^2}$

Total 4x4 Array Multiplier Active Area:

- 16 AND GATES: $16 * 3.375 = \mathbf{54 \mu m^2}$
- 7 HALF ADDER: $7 * 6.750 = \mathbf{47.25 \mu m^2}$
- 9 FULL ADDERS: $9 * 6.750 = \mathbf{60.75 \mu m^2}$

TOTAL SILICON AREA: 192.375 μm^2

POWER CONSUMPTION ANALYSIS:

$$P_{tot} = P_{dyn} + P_{sc} + P_{stat}$$

$$P_{dyn} = \alpha * C_{load} * V_{dd}^2 * f$$

$$P_{sc} = I_{sc} * V_{dd} * f$$

$$P_{stat} = I_{leak} * V_{dd}$$

1. Power Consumption by TGXOR Gate:

$$P_{avg} = 13.918 \mu W$$

$$P_{stress} = 120 \mu W$$

2. Power Consumption by HALF ADDER:

$$P_{avg} = 90.19 \mu W$$

$$P_{stress} = 126.847 \mu W$$

3. Power Consumption by FULL ADDER:

$$P_{avg} = 192.312 \mu W$$

$$P_{stress} = 708.537 \mu W$$

4. Power Consumption by RIPPLE ADDER:

$$P_{avg} = 100.98 \mu W$$

$$P_{stress} = 306.3 \mu W$$

5. Power Consumption by MULTIPLIER:

$$P_{avg} = 762.211 \mu W$$

$$P_{stress} = 3.273 mW$$

COMPARISON BETWEEN CMOS MULTIPLIER AND OPTIMISED MULTIPLIER:

Metric	Interim Report (Unoptimized)	Final Report (Optimised)
<u>Critical Path Delay</u>	2.58 ns	1.0443 ns
<u>Total Active Area</u>	439.5 μm^2	192.375 μm^2
<u>Multiplier Architecture</u>	16 ANDs + 12 Full Adders	16 ANDs + 7 Half Adders + 9 Full Adders
<u>Full Adder Topology</u>	Standard Static CMOS (34 Transistors)	Hybrid TG-XOR (18 Transistors)
<u>Ripple Adder Topology</u>	4 Full Adder based on Static CMOS transistors	1 HA and 3 FA based on Hybrid TGXOR Transistors

CONTRIBUTION BY EACH INDIVIDUAL:

- VARUN HIRANI (2304241)
 - Designed the baseline schematics for all the standard logic gates (AND, OR, NOT, XOR, FULL ADDER, 4-BIT ADDER, and Array Multiplier (CMOS STATIC)).
 - Handled the voltage transfer charts, timing diagrams, power consumption, area calculation, and initial delay measurements for the first report.
 - Explored different transistor-level designs to find better options to replace the standard gates.
- KAMAL ASSUDANI (2306317)
 - Co-worked with Varun in generating the timing diagrams for the interim report.
 - Took charge of drawing the complex, custom schematics for the second half of the project (the 6T TG-XOR, the optimised adders, and the final 4x4 multiplier matrix).
 - Worked closely with Varun to review the researched designs and select the best for our project.
 - Handled the final dynamic stress tests, measured the peak and average power usage, and calculated the total silicon area for the final report.

REFERENCES:

Literature Reviewed for Transistor-Level Optimisation (TSERF & TG-Logic)

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- Zhuang, N., & Wu, H. (1992). A new CMOS full-adder design. *IEEE Journal of Solid-State Circuits*, 27(5), 840-844.
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Web Resources & Schematic References

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- ResearchGate. Proposed SERF full adder circuit [Figure 2]. Retrieved from https://www.researchgate.net/figure/Proposed-SERF-full-adder-circuit_fig2_221381046

Software & Simulation Models

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- Predictive Technology Model (PTM). *Generic 250nm CMOS Process Models*.